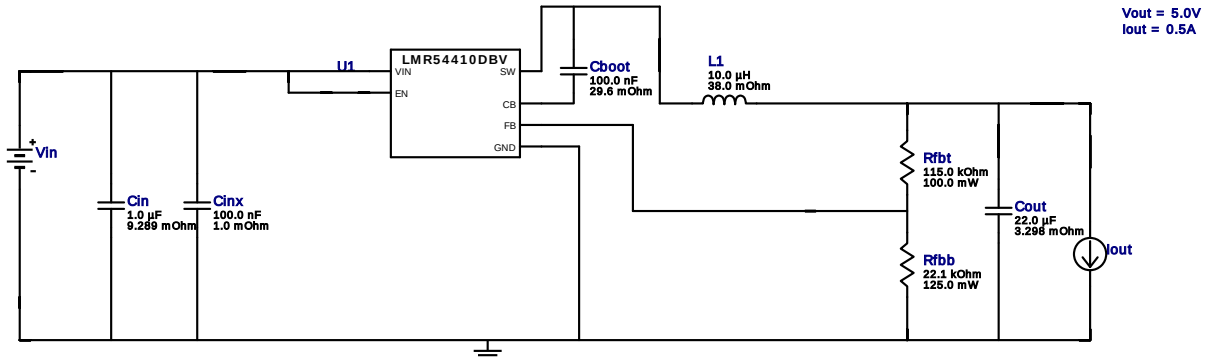


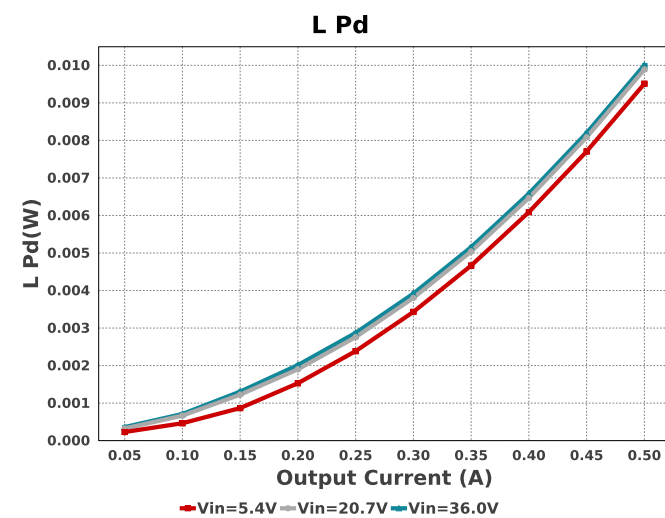
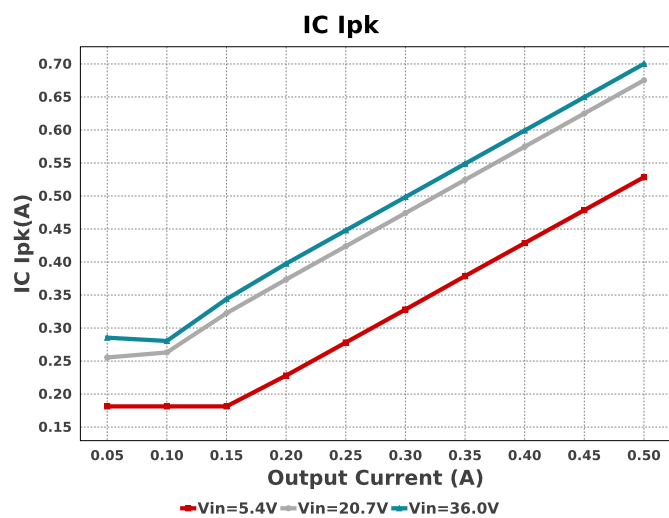
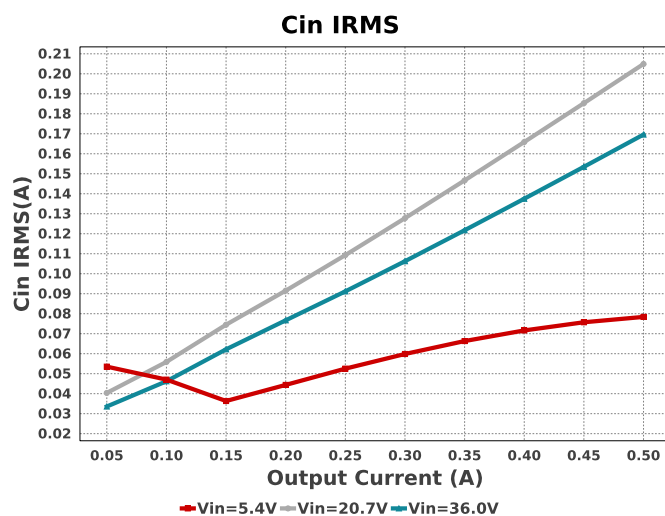
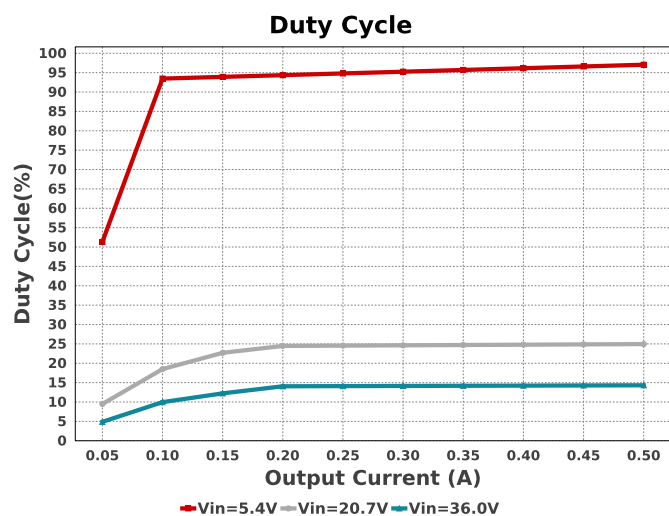
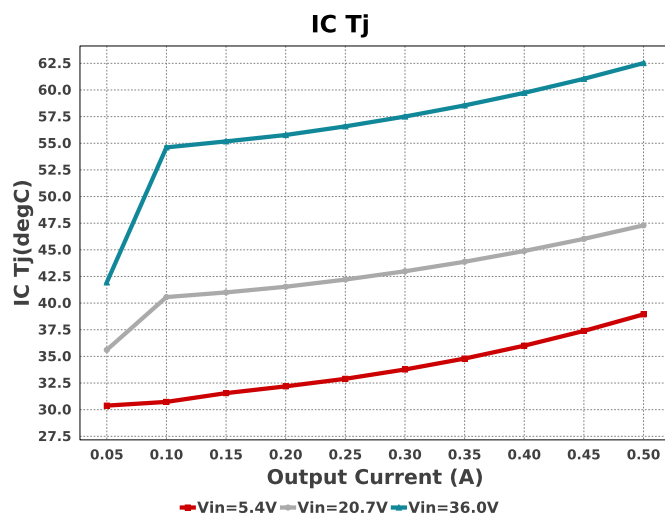
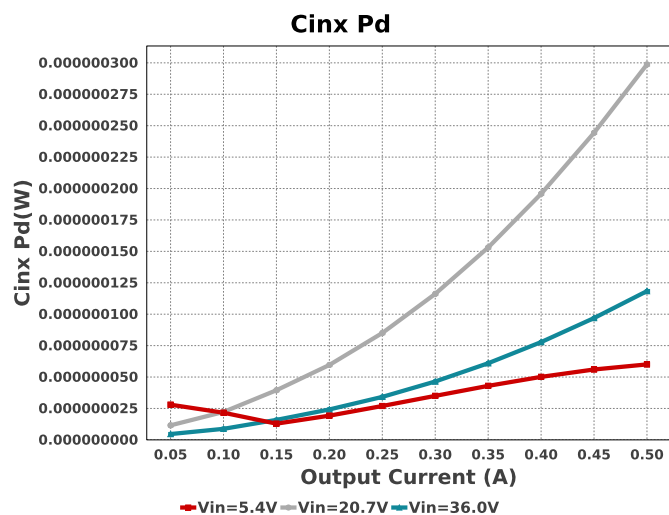
WEBENCH® Design Report

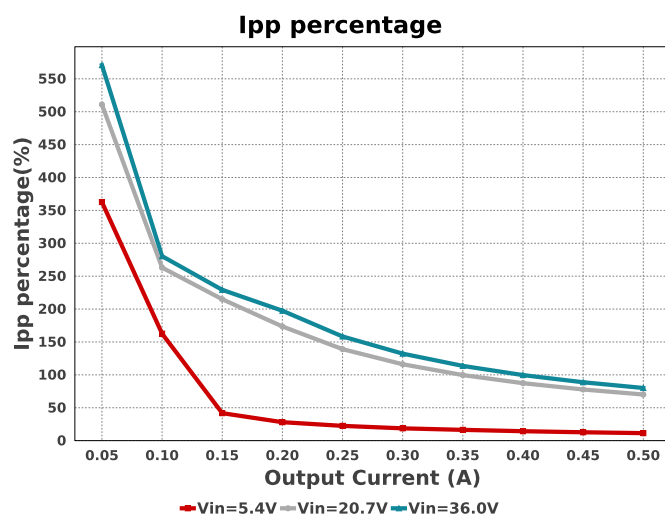
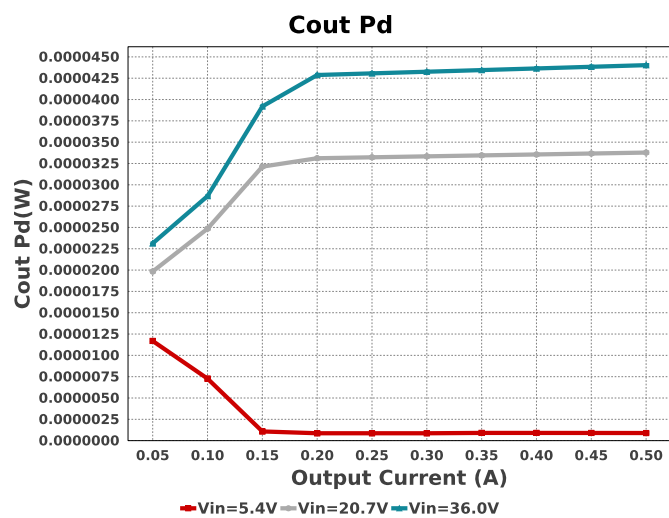
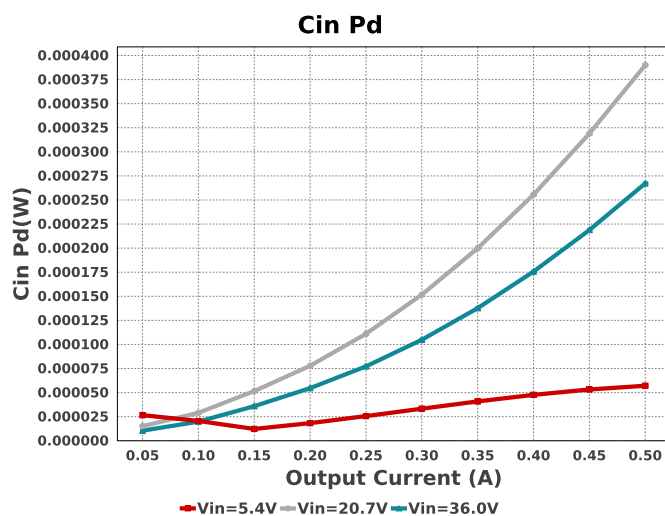
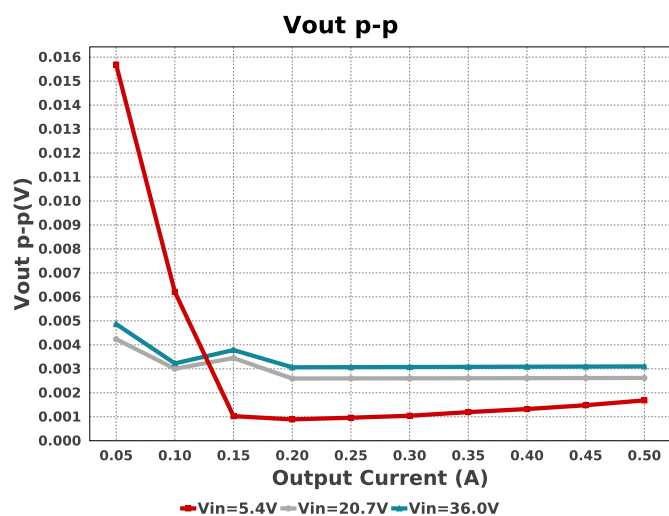
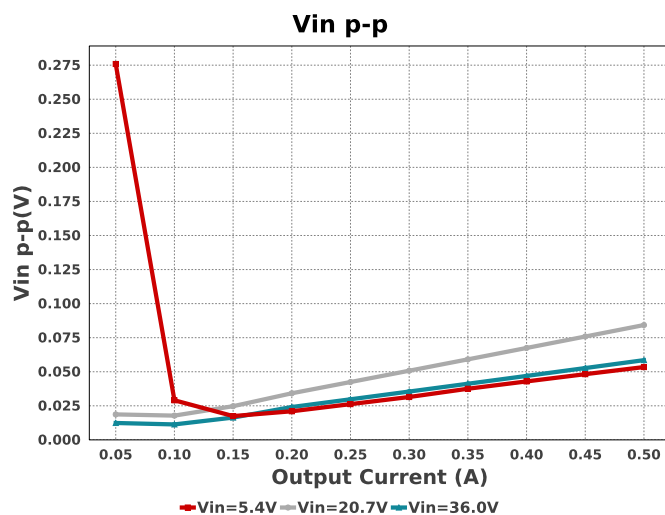
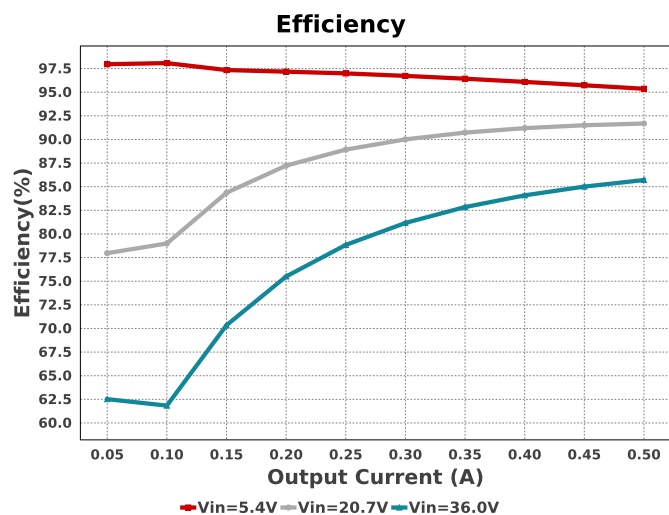
Design : 47 LMR54410DBVR
LMR54410DBVR 5.4V-36V to 5.00V @ 0.5A GBG-PCB V2.3

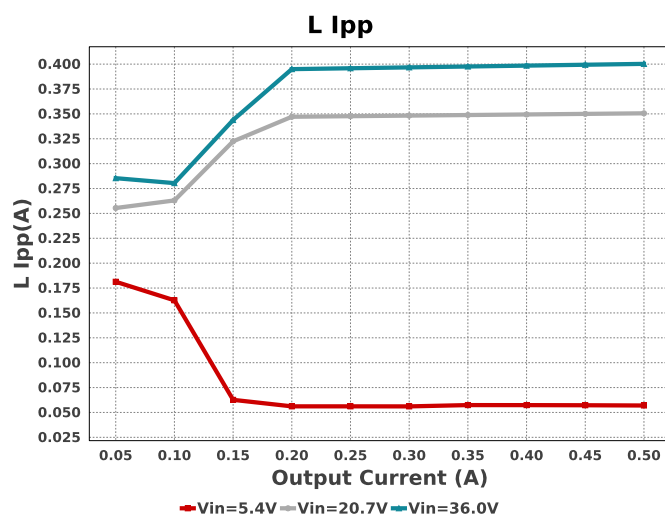
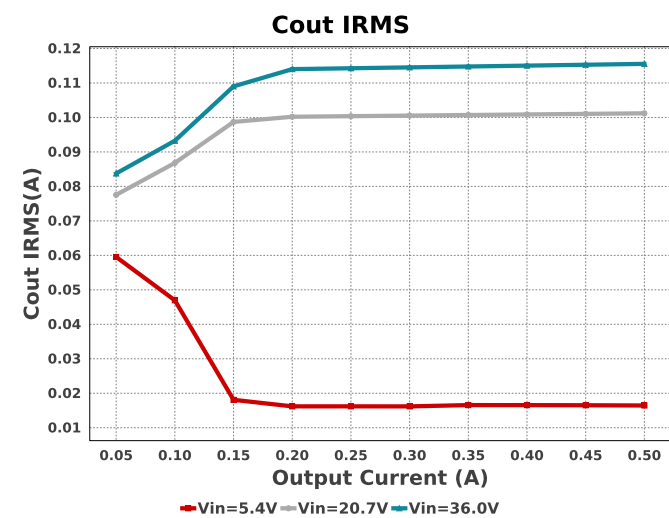
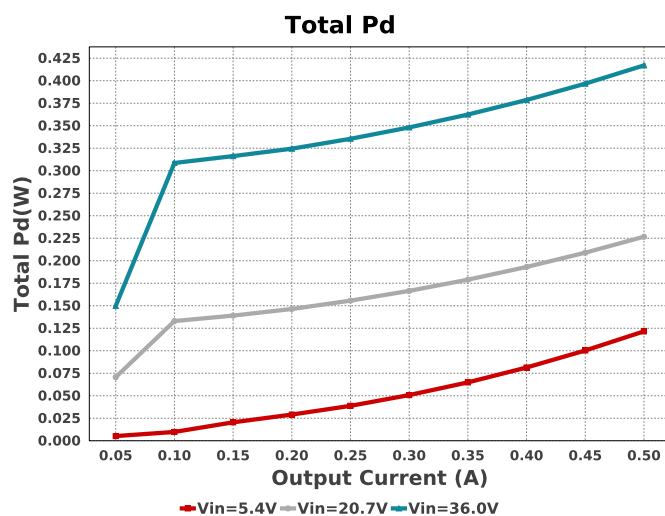
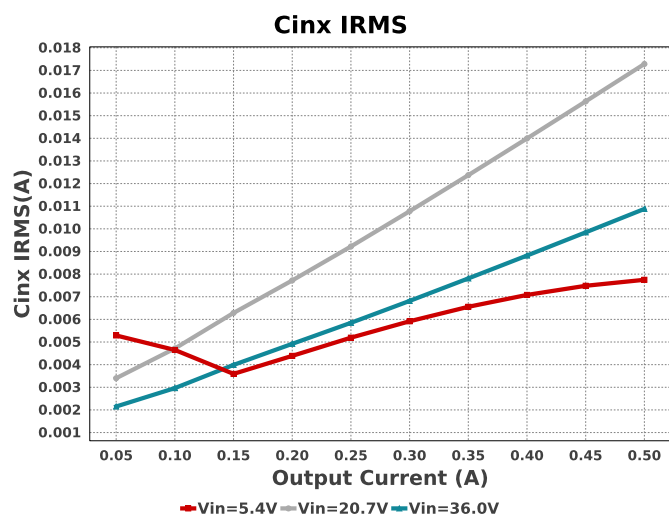
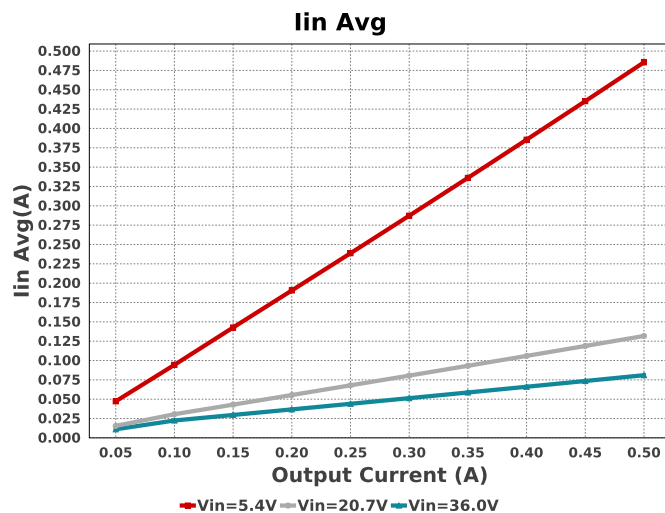
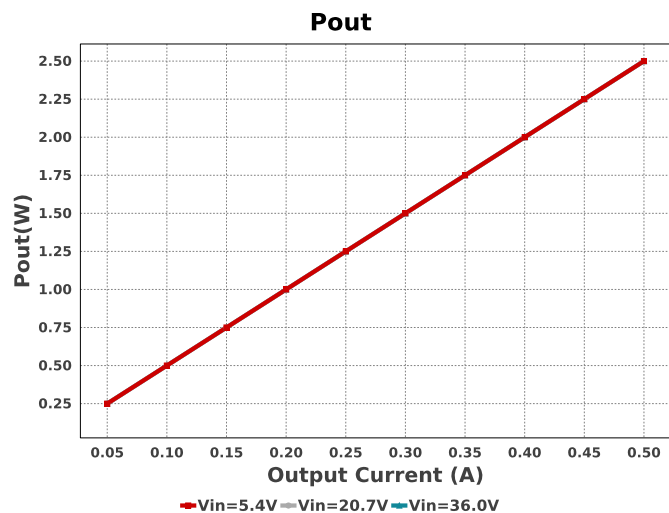


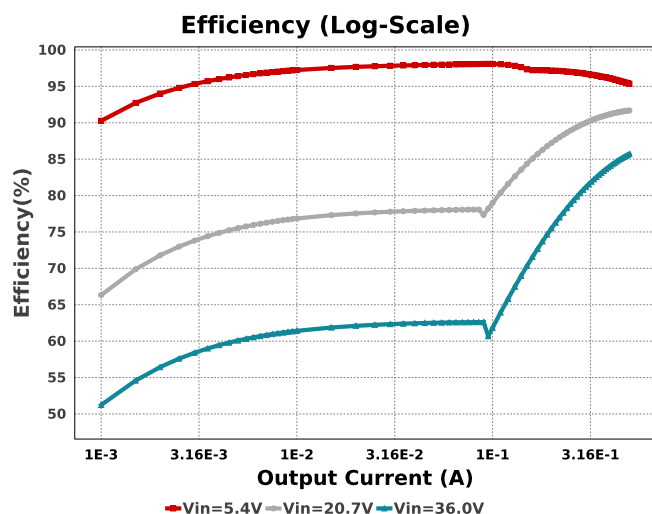
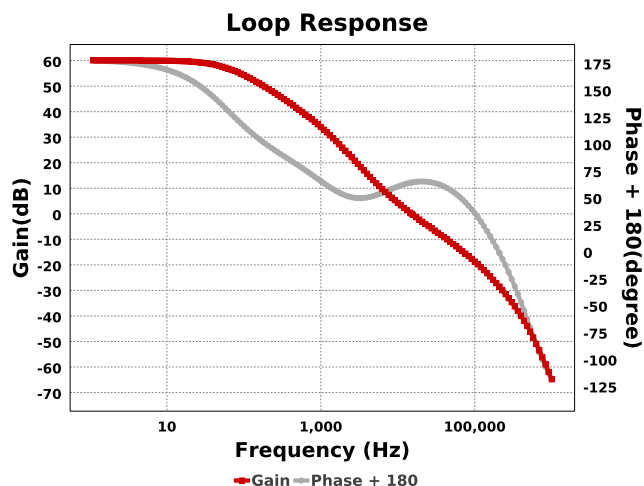
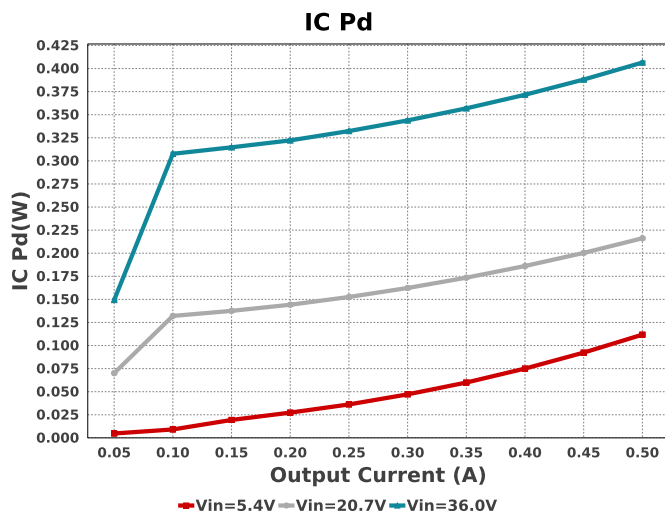
Electrical BOM

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	TDK	CGA3E2X7R1E104K080AA Series= X7R	Cap= 100.0 nF ESR= 29.6 mOhm VDC= 25.0 V IRMS= 0.0 A	1	\$0.01	 0603 5 mm ²
Cin	TDK	C3216X7R1H105K160AB Series= X7R	Cap= 1.0 uF ESR= 9.289 mOhm VDC= 50.0 V IRMS= 2.38446 A	1	\$0.06	 1206_180 11 mm ²
Cinx	MuRata	GRM188R72A104KA35D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 100.0 V IRMS= 3.85 A	1	\$0.04	 0603 5 mm ²
Cout	Taiyo Yuden	MSASE32MSB7226KPNB25 Series= X7R	Cap= 22.0 uF ESR= 3.298 mOhm VDC= 16.0 V IRMS= 3.7481 A	1	\$0.15	 1210 15 mm ²
L1	Bourns	SRR6038-100Y	L= 10.0 uH 38.0 mOhm	1	\$0.39	 SRR6038 77 mm ²
Rfbb	Vishay Dale	CRCW080522K1FKEA Series= CRCW..e3	Res= 22.1 kOhm Power= 125.0 mW Tolerance= 1.0%	1	\$0.04	 0603 2 mm ²
Rfbb	Vishay Dale	CRCW0805115KFKEA Series= CRCW..e3	Res= 115.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.04	 0603 2 mm ²
U1	Texas Instruments	LMR54410DBVR	Switcher	1	\$0.36	 DBV0006A 15 mm ²









Operating Values

#	Name	Value	Category	Description
1.	BOM Count	8		Total Design BOM count
2.	Total BOM	\$1.102		Total BOM Cost
3.	Cin IRMS	169.61 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	267.22 μ W	Capacitor	Input capacitor power dissipation
5.	Cinx IRMS	10.88 mA	Capacitor	Bulk capacitor RMS ripple current
6.	Cinx Pd	118.37 nW	Capacitor	Bulk capacitor power dissipation
7.	Cout IRMS	115.545 mA	Capacitor	Output capacitor RMS ripple current
8.	Cout Pd	44.031 μ W	Capacitor	Output capacitor power dissipation
9.	IC Ipk	700.131 mA	IC	Peak switch current in IC
10.	IC Pd	406.52 mW	IC	IC power dissipation
11.	IC Tj	62.521 degC	IC	IC junction temperature
12.	IC Tolerance	20.0 mV	IC	IC Feedback Tolerance
13.	ICThetaJA Effective	80.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
14.	Iin Avg	81.028 mA	IC	Average input current
15.	Ipp percentage	80.052 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
16.	L Ipp	400.261 mA	Inductor	Peak-to-peak inductor ripple current
17.	L Pd	10.007 mW	Inductor	Inductor power dissipation
18.	Cin Pd	267.22 μ W	Power	Input capacitor power dissipation
19.	Cinx Pd	118.37 nW	Power	Bulk capacitor power dissipation
20.	Cout Pd	44.031 μ W	Power	Output capacitor power dissipation
21.	IC Pd	406.52 mW	Power	IC power dissipation
22.	L Pd	10.007 mW	Power	Inductor power dissipation
23.	Total Pd	417.015 mW	Power	Total Power Dissipation
24.	Cross Freq	15.164 kHz	System	Bode plot crossover frequency
25.	Duty Cycle	14.315 %	System	Duty cycle
26.	Efficiency	85.704 %	System	Steady state efficiency
27.	FootPrint	141.0 mm ²	System	Total Foot Print Area of BOM components

#	Name	Value	Category	Description
28.	Frequency	1.1 MHz	System Information	Switching frequency
29.	Gain Marg	-28.606 dB	System Information	Bode Plot Gain Margin
30.	Iout	500.0 mA	System Information	Iout operating point
31.	Low Freq Gain	60.061 dB	System Information	Gain at 1Hz
32.	Mode	CCM	System Information	Conduction Mode
33.	Phase Marg	65.561 deg	System Information	Bode Plot Phase Margin
34.	Pout	2.5 W	System Information	Total output power
35.	Vin	36.0 V	System Information	Vin operating point
36.	Vin p-p	58.506 mV	System Information	Peak-to-peak input voltage
37.	Vout	5.0 V	System Information	Operational Output Voltage
38.	Vout Actual	4.963 V	System Information	Vout Actual calculated based on selected voltage divider resistors
39.	Vout Tolerance	4.237 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
40.	Vout p-p	3.1 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	500.0 m	Maximum Output Current
VinMax	36.0	Maximum input voltage
VinMin	5.4	Minimum input voltage
Vout	5.0	Output Voltage
base_pn	LMR54410	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of L1 before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

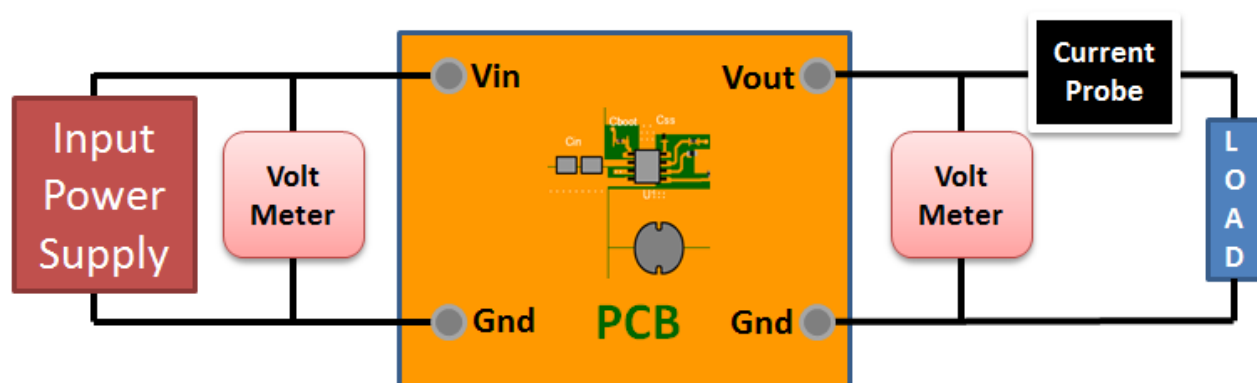
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 5.4V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 36B639548DBC54FC6F0179AABB8E44BC[v1]
2. **LMR54410** Product Folder : <http://www.ti.com/product/LMR54410> : contains the data sheet and other resources.

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